



IGL BBS User Guide

Behaviour Based Safety — observation system

Everything you need to record a BBS observation, track corrective actions, read the dashboards and switch on the AI assistant. Written for every department — no software experience needed.

Version 1.1.0 · India Glycols Limited, Kashipur

Who is this for

Observers, HODs,
EHS and plant leadership

Time to first observation

Under 3 minutes

Works on

Desktop, tablet
and mobile — even offline

IGL BBS replaces the paper BBSO checklist. You record what you saw during a safety observation, the app does the counting, builds the trends, and creates corrective actions automatically. Everything is open to **all departments**.

1 Open the app

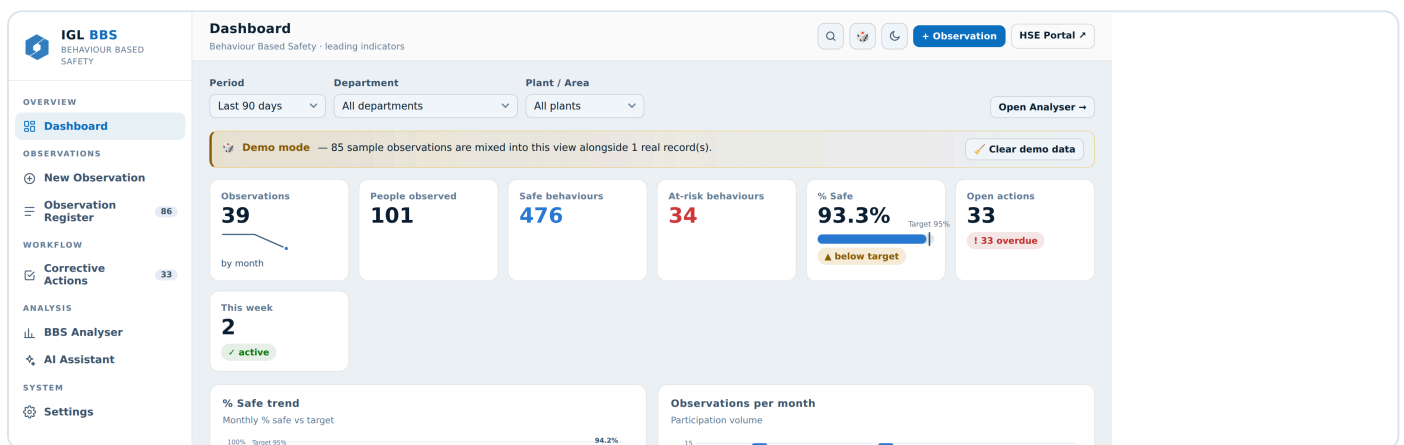
Go to the BBS web address shared by EHS. There is no login and nothing to install.

2 Set your defaults once

Settings → *My defaults* — your name, department and plant. Every new observation is pre-filled from then on.

3 Try it with sample data first

Click the  button at the top right and choose *Load 60 records*. You can explore every screen safely, then clear it in one click.



The main screen. Left: the menu. Top right: search, demo data, light/dark theme and the + *Observation* button. Middle: your safety numbers for the selected period.

The menu at a glance

Screen	What you use it for
Dashboard	The headline numbers — % safe, trends, where at-risk behaviour is concentrated, overdue actions
New Observation	Record an observation using the IGL BBSO checklist
Observation Register	Search, open, print and export every record
Corrective Actions	Track and close the actions raised from at-risk findings
BBS Analyser	Deeper analysis by category, month, department and observer
AI Assistant	Insights, toolbox talks, reports and a safety chat (needs a free API key — page 8)
Settings	Your defaults, targets, checklist editor, backups and demo data

Tip — the fastest way to move around

Press **Ctrl** + **K** anywhere to search observations, actions and checklist items, or to jump to any screen. Press **?** to see all shortcuts.

Set the three filters at the top — **Period**, **Department** and **Plant/Area** — and every number, chart and table below changes to match.



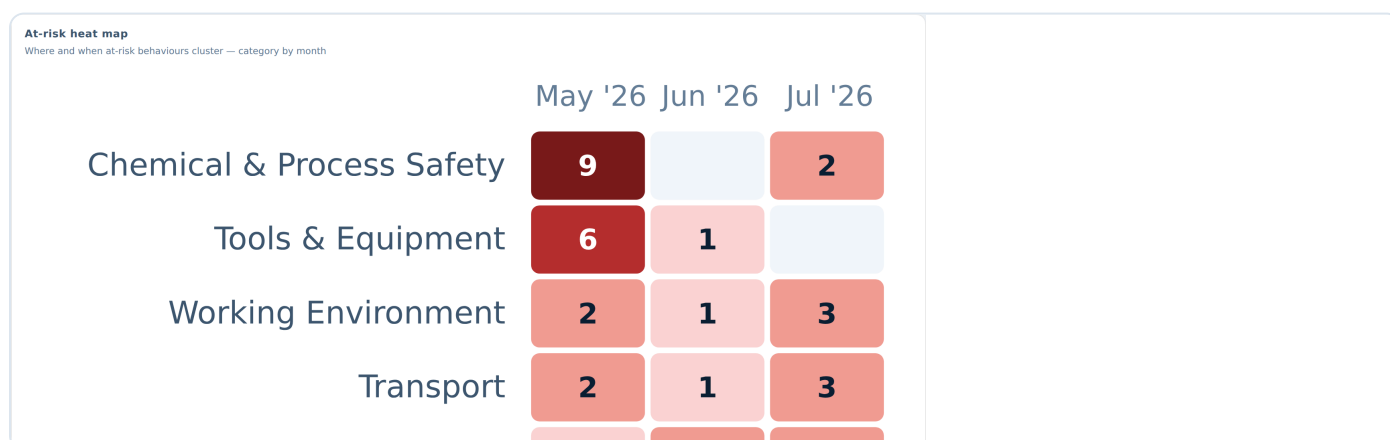
The tiles. Safe and at-risk counts, **% Safe** against the target, open and overdue actions, plus this week's participation and your reporting streak.

How % Safe is calculated — exactly as in the Excel workbook

$\% \text{ Safe} = \text{Total Safe} \div (\text{Total Safe} + \text{Total At Risk}) \times 100$ · Green = on or above target, amber = 80-95%, red = below 80%. Set the target in Settings.



Trends. Left: % safe each month against target. Right: how many observations were recorded — participation. Hover any bar or point for exact numbers.



The heat map shows where and when at-risk behaviour clusters. The darkest cells are where to focus next month. Below it, the app lists departments that recorded nothing this period.

Click + **Observation** at the top right (or press **N**). Fill the header, then work through the checklist.

Observation details
As per IGL BBSO format

Observer name *	Department *	Plant / Unit *	Location *	Date *
Brajraj Singh	Production – MEG / Glycc ▾	MEG ▾	Analyser House, MEG	07/30/2026 📅
Time	Shift	Observation type	People observed	
09:10 PM ⌚	General ▾	Planned ▾	1	
Job / activity description *				
Carrier gas cylinder change on selectivity GC				

Observation details. Fields marked * are required. Date, time and shift default to now; your name, department and plant come from your saved defaults.

Three ways to fill the checklist

① Tap the counters

The normal way — press + under **SAFE** or **RISK** for each item you observed.

② Section shortcut

Hover a section heading and click ✓ **All safe** to mark the whole section at once, then correct the exceptions.

③ Let AI do it

Type what you saw in plain words and the AI marks the checklist for you. You review before saving.

+ AI quick fill Describe the observation in plain words — AI marks the checklist for you (review before saving)

Two fitters replacing a pump seal at MEG. Both wearing full PPE and the work permit was displayed. One was standing under a suspended load for a moment and there was oil spillage near the walkway.

+ Fill checklist with AI 🗣️ **Scan attached photos** 📷

AI quick fill (optional). Describe the observation the way you would tell a colleague, then click *Fill checklist with AI*. Use the 🗣️ button to speak instead of typing. This needs the API key from page 8.

Always review what the AI marked

The AI is an assistant, not the observer. Check every marked item and remark before you save — you remain responsible for the record.

Photos

Scroll to the **Photos** card to attach up to four site photos. On a phone this opens the camera directly. Images are automatically resized before saving so they do not fill up storage.

The checklist has **11 sections and 78 items** — the same IGL BBSO format as the paper sheet. Click a section heading to open it. Use the **Find an item** box to jump straight to an item.

▼ 2. Personal Protective Equipment (PPE) Safe 3 Risk 1

2.1 Wears required PPE SAFE - 0 + RISK - 1 +
 Helmet chin strap not worn properly. Corrected at site immediately. ✓ Corrected on spot Create action ⚠ High risk

2.2 PPE complete and properly worn SAFE - 2 + RISK - 0 +
 Remark (what was seen, correction done...)

2.3 Uses task specific PPE SAFE - 1 + RISK - 0 +
 Remark (what was seen, correction done...)

2.4 PPE in good condition SAFE - 0 + RISK - 0 +

2.5 Uses special PPE in hazardous areas SAFE - 0 + RISK - 0 +

Marking an item. Increase **SAFE** for correct behaviour, **RISK** for an at-risk one. As soon as you mark something, a remark box appears.

When you mark an item At Risk

- **Write a remark** — what you saw and what you did about it. Use to dictate.
- ✓ Corrected on spot — you fixed it there and then.
- ⚙ Create action — raises a corrective action automatically (on by default).
- ⚠ High risk — shortens the action due date from 7 days to 3.

Total Safe 4 Total At Risk 1 % Safe 80% Feedback discussed with person(s) Reset Save observation

The bar at the bottom stays with you and totals everything live — Total Safe, Total At Risk and % Safe, exactly as the Excel sheet calculated them.

1 Confirm the feedback toggle

Tick *Feedback discussed with person(s)* — the conversation is the point of BBS, not the paperwork.

2 Save

Click **Save observation** (or press **Ctrl + S**). You get an ID like **BBS-0042**.

3 Check the corrective actions

The app tells you how many were created. They appear under *Corrective Actions* straight away.

Progress ring

The circle at the top of the checklist shows how many of the 78 items you have marked. There is no need to mark all of them — record only what you actually observed.

Every saved record lives here. Search by ID, observer, location or job; filter by department, plant and date range; click any column heading to sort.

Observation Register									
All BBS observation records									
☐	0055	30 Jul 26	P. Joshi	Electrical Maintenance	Tank Farm	Catalyst charging	19	0	✓ 100%
☐	BBS-0024	30 Jul 26	S. Chauhan	Laboratory / QC	Tank Farm	Filter cleaning	19	0	✓ 100%
☐	BBS-0084	29 Jul 26	M. Ansari	Production - MEG / Glycols	Warehouse	Hot work — welding	11	1	▲ 91.7%
☐	BBS-0034	28 Jul 26	M. Ansari	EHS / Fire & Safety	Boiler House	Cable laying	15	2	▲ 88.2%
☐	BBS-0047	25 Jul 26	T. Rana	Instrumentation	Glycol Ethers	Catalyst charging	12	2	▲ 85.7%
☐	BBS-0033	20 Jul 26	R. Verma	Production - Specialty Chemicals	Ethoxylates	Filter cleaning	8	1	▲ 88.9%
☐	BBS-0007	18 Jul 26	K. Pandey	Power Plant & Utilities	Ethoxylates	Scaffold erection	7	1	▲ 87.5%
☐	BBS-0031	17 Jul 26	T. Rana	HR & Administration	Distillery / ENA	Height work — pipe rack	15	0	✓ 100%
☐	BBS-0001	16 Jul 26	Brajraj Singh	Production - MEG / Glycols	MEG	Carrier gas cylinder of sel...	7	1	▲ 87.5%
Loading...									

Register. Tick the boxes on the left to select several records, then export or delete them together. A deletion can always be undone from the message that appears.

Click a row to open the full record

- **Print** — a clean BBSO sheet with signature lines (also use it to save a PDF)
- **Excel (BBSO)** — the record in the original workbook layout
- **Duplicate** — start a new observation from this one (handy for routine rounds)
- **Edit / Delete** — correct or remove the record
- **AI actions, Root cause, Severity** — see pages 8–9

Observation BBS-0051

Observer

Department

Plant / Unit

Location

Date & time

Shift

Job description

Type

People observed

Feedback given

M. Ansari

Mechanical Maintenance

ETP

Area 1

27 Jul 26 - 14:00

General

Filter cleaning

Planned

3

No

REF	ITEM	SAFE	AT RISK	REMARKS
2.1	Wears required PPE	1		
2.5	Uses special PPE in hazardous areas	2		
3.7	Safe compressed air use	2		
3.10	Containers/drums safe	2		
5.3	Proper labelling/storage	2		
5.6	Process within limits	2		
5.9	Reaction/pressure control	1		
5.10	Safe sampling/blending	1		
6.6	HRA/SA followed		1	Barricading missing around excavation - ✓ corrected on spot - > high risk
7.1	Reports unsafe conditions	1		
Total Safe 14		Total At Risk 1		% Safe ▲ 93.3%

Print

Excel (BBS0)

All actions

Root cause

Severity

Duplicate

Edit

Delete

Full record with every marked item and remark.

Exports

Button	What you get
↓ Excel	The whole filtered register as .xlsx, with a second sheet of corrective actions
↓ CSV	The same data as a plain CSV file
↓ Backup	A .json file with <i>everything</i> — use this to move or merge data between computers
↑ Import	Load a backup — choose Merge to add to what is already there, or Replace to overwrite

How departments and EHS share data

Each computer keeps its own records. Once a week each department clicks **↓ Backup** and sends the file to EHS; EHS clicks **↑ Import → Merge** to build the consolidated plant view.

Corrective Actions									
CAPA arising from BBS observations									
☐	0017	! Overdue	of drums	BBS-0014	Ethoxylates	A. Sharma	26	High	
☐	ACT-0012	! Overdue	[1.7] Avoids cramped/unstable positions — Tool without inspection tag	BBS-0010	Stores & Purchase	K. Pandey	09 Feb 26	High	
☐	ACT-0063	! Overdue	[10.6] Transport compliance followed — Working without valid PTW copy at site	BBS-0078	HR & Administration	S. Chauhan	09 Feb 26	Medium	
☐	ACT-0005	✓ Closed	[10.3] Loads secured — Chin strap loose, corrected immediately	BBS-0004	Instrumentation	K. Pandey	10 Feb 26	High	
☐	ACT-0010	! Overdue	[4.3] Walkways clear — Full body harness not anchored	BBS-0009	Boiler House	Brajraj Singh	13 Feb 26	Medium	
☐	ACT-0013	! Overdue	[3.6] Instruments functional — PPE not worn properly, corrected at site	BBS-0010	Stores & Purchase	P. Joshi	13 Feb 26	Medium	
☐	ACT-0011	✓ Closed	[5.3] Proper labeling/storage — Gas cylinder not chained	BBS-0009	Boiler House	Brajraj Singh	13 Feb 26	Medium	
☐	ACT-0003	✓ Closed	[10.1] Safe loading/unloading — Chin strap loose, corrected immediately	BBS-0004	Instrumentation	P. Joshi	14 Feb 26	Medium	
☐	ACT-0004	✓ Closed	[3.9] Safe tanker/loading connections — Housekeeping poor near pump area	BBS-0004	Instrumentation	K. Pandey	14 Feb 26	Medium	
☐	ACT-		[R.11] Uses fall protection — Tool without inspection				16 Feb		

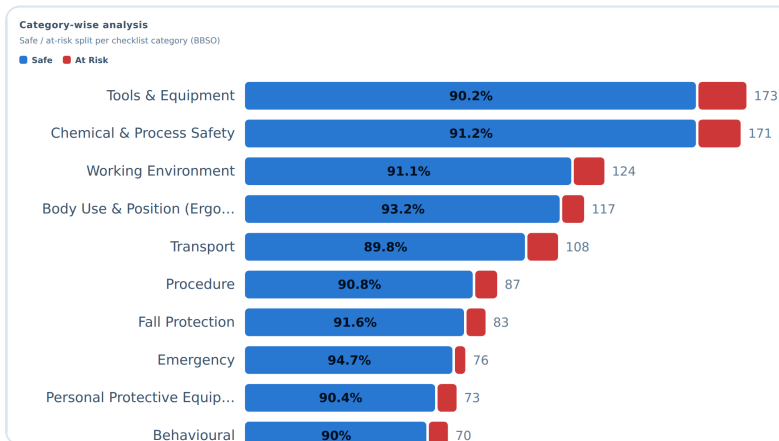
Corrective actions. Created automatically from at-risk findings — **High** priority items are due in 3 days, others in 7.

Working through them

- Click a row to assign an owner, change the due date or close it with a verification remark
- Tick several rows to **close**, **assign** or **delete** them together
- Filter by **Overdue** to see what needs attention today
- Status: ! Open ● In Progress ✓ Closed ! Overdue

Analyser — the deeper view

- Category-wise analysis** — safe vs at-risk for all 11 sections, with a totals row
- Monthly trend** — % safe by month with the same numbers as a table
- Department scorecard** — participation against the monthly target
- Observer leaderboard** — who is active, and who is finding at-risk behaviour
- ↓ Analyser report** — a six-sheet Excel workbook of all of it



BBS Analyser. Filter by period, department, plant or observer — every chart and table below follows the filter.

The app works fully without AI. To unlock the AI features you add a free **OpenRouter** key once, in your own browser. It takes about two minutes.

1 Create a free OpenRouter account

Open `openrouter.ai` and sign up — Google sign-in works.

2 Create a key

Go to `openrouter.ai/keys` → *Create key* → name it “IGL BBS” → copy it. It starts with `sk-or-v1-`.

3 Paste it into the app

AI Assistant → **Setup & models** → paste into the *API key* box. It saves as you type.

4 Test it

Click **Test connection** — you should see ✓ Connected.

5 Pick your models

Click **Load live model list**, then choose a **Primary model**. Anything ending in `:free` costs nothing.

OpenRouter connection
One key unlocks hundreds of models — many free. Get one at openrouter.ai/keys

API key (stored only in this browser)

sk-or-v1-...

Show

Connection

Test connection

Primary model

meta-llama/llama-3.3-70b-instruct:free · free

Fallback (used when the primary is rate-limited)

— none —

Vision model (photo hazard scan)

google/gemini-2.5-flash

Load live model list

Stream responses live

AI Assistant → **Setup & models**. Paste the key on the left, test on the right, then choose the three models underneath.

What the three model boxes do

Setting	Purpose	Suggested
Primary model	Everything — insights, reports, chat	<code>llama-3.3-70b-instruct:free</code>
Fallback	Used automatically if the primary is busy	A second <code>:free</code> model
Vision model	Reads site photos for the hazard scan	<code>google/gemini-2.5-flash</code>

If something does not work

401 wrong key — paste it again · **402** no credit — switch to a `:free` model · **429** too many requests — wait a minute or set a fallback.

 Data insights Patterns, risks and next-month focus areas from the last 90 days	 Toolbox talk A 5-minute TBT script on your current top at-risk behaviour	 Root cause (5-Why) Pick an observation and get a 5-Why plus contributing factors
 CAPA review Which open actions to close first, duplicates, verification steps	 Weekly digest A 60-second leadership brief comparing this week with last	 Safety quiz 10 MCQs built from your own at-risk findings, with answer key
 Photo hazard scan Upload a site photo — the AI lists hazards and maps them to the checklist	 Severity scoring Score an observation's findings and re-prioritise its corrective actions	 Checklist auto-fill Describe what you saw in plain words — the AI marks the BBSO checklist

AI Assistant → AI tools. Click any card and the answer is written live on screen. Every tool works on the data already in your browser.

 Data insights Patterns, risks needing management attention, and three focus areas for next month.	 Toolbox talk A ready 5-minute TBT script on your top at-risk behaviour — read it at the shift briefing.
 Root cause (5-Why) Pick an observation and get a 5-Why chain plus contributing factors and two preventive barriers.	 CAPA review Which open actions to close first, which are duplicates, and how to verify closure.
 Weekly digest A 60-second brief for the plant head comparing this week with last.	 Safety quiz Ten multiple-choice questions built from your own findings, with an answer key — use it in training.
 Photo hazard scan Upload a site photo; the AI lists what is unsafe and maps it to checklist items.	 Severity scoring Scores findings for severity and likelihood, then re-prioritises the linked actions.

Hindi and Hinglish

Any generated text can be translated with one click using the **अ हिंदी** or **Hinglish** button, so it can be read out on the shop floor.

Is our data safe?

The key and all observations stay in your browser. Only summarised statistics — or the one observation you picked — go directly to OpenRouter. Nothing passes through a company server.

Safety assistant chat
 Knows your last 90 days of BBS data. Ask about trends, standards, PPE, PTW, LOTO, toolbox topics...

Namaste! I am your BBS safety assistant. Ask me about your observation data, or any safety topic — PPE, work permits, LOTO, work at height, chemical handling...

Which department needs most attention? Draft a toolbox talk on line of fire Top 3 focus areas for next month Explain our % safe trend to a plant head

Safety chat knows your last 90 days of data. Ask “which department needs most attention?” or any safety question. Save your own prompts in Setup to reuse them.

Monthly BBS report

A formal report drafted from that month's data — ready to circulate to HODs and plant leadership

Jul '26

+ Generate report

AI Assistant → Reports. Choose a month, click *Generate report*, then **Copy**, **Download** or **Print / PDF** it for circulation to HODs.

Settings you will actually use

- **My defaults** — name, department, plant (set once)
- **Targets** — % safe target and monthly observations per department
- **Appearance** — light or dark theme, compact or comfortable rows
- **Departments / Plants** — edit the dropdown lists, one per line
- **Checklist editor** — reword an item, switch one off, or add your own under *Others*
- **Data management** — backup, import, demo data and a storage meter

Demo data

Sample observations let you explore every dashboard, chart and report before real data exists. Demo records are tagged — clearing them never touches records you entered yourself.

86	85	1	63
observations total	of them demo	entered by you	corrective actions

Load 30 records

Load 60 records

Load 200 records

Clear 85 demo records

Records span the last 6 months with a slowly improving safety trend, so the charts show a realistic story.

Close

Demo data (the button, top right). Load 30, 60 or 200 sample records.

Demo data is for testing only

Sample records are clearly tagged and a yellow *Demo mode* bar appears on every screen while they exist. **Clear demo data** removes only those records — anything you entered yourself is untouched. Always clear it before the app goes live in your department.

Data management

Data is stored in this browser (offline-capable). Use backups to move or merge data between computers — e.g. departments export weekly and EHS imports & merges.

Using 0.08 MB of roughly 5 MB browser storage · 86 observations · 0 photos

↓ Export backup (JSON)

↑ Import

↓ Export register (Excel)

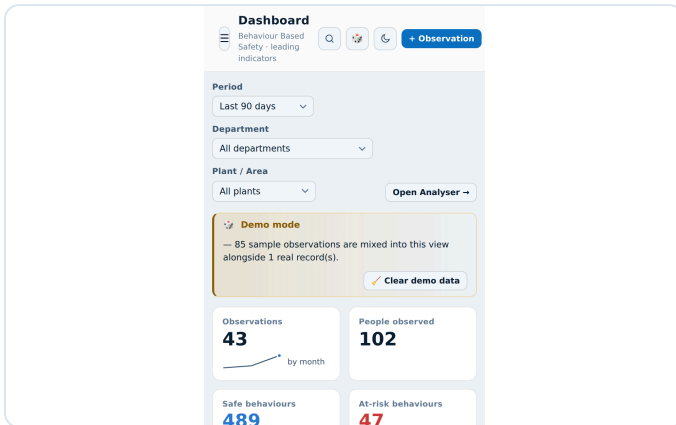
Load 60 demo records

Load 200

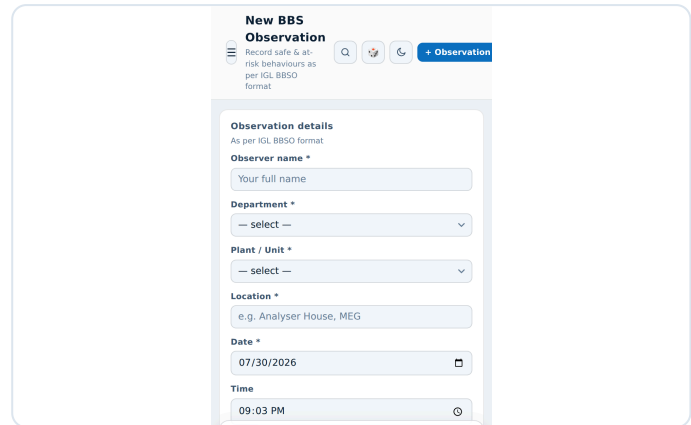
Clear demo data (85)

Clear all data

Settings → Data management. The meter shows how much browser storage is used. Export a backup regularly — it is the only copy of your data.



Mobile. Tabs along the bottom for the screens you use most.



Recording in the field — camera, dictation and the running totals all work.

Install it like an app

On the phone open the site, then use the browser menu → **Add to Home Screen**. It opens full-screen and keeps working without network — your observations save on the phone and are there when you come back into coverage.

Keyboard shortcuts (desktop)

Keys	Action
Ctrl + K	Search everything and jump anywhere
N	Start a new observation
G then D / R / A / Y	Go to Dashboard / Register / Actions / Analyser
Ctrl + S	Save the observation you are writing
T · ? · Esc	Theme · all shortcuts · close a dialog

Common questions

Question	Answer
Where is my data stored?	In your own browser on this computer or phone. Nothing is sent to a server. Take a backup regularly.
Can two people share one register?	Not automatically — each person exports a backup and EHS merges them.
Do I need the AI?	No. Every core feature works without a key. The AI only adds analysis and drafting.
Does the AI cost money?	Not with a : free model. Usage is shown under AI Assistant → Usage .
I deleted a record by mistake	Click Undo in the message that appears. If it has gone, restore your last backup.

Need help?

Contact the EHS department, Kashipur. This app is part of the **IGL HSE Portal** — the link is at the bottom of the menu.